

Science Practice

SCIENCE UNIT TEST COMPLETE REVISION PAPER

Chapters:

- Ch-1 Matter in Our Surroundings
- Ch-5 The Fundamental Unit of Life
- Ch-7 Motion

SECTION A

MCQs (15 × 1 = 15)

- Which scientist first observed living cells?
 - Hooke
 - Virchow
 - Leeuwenhoek
 - Schwann
- The SI unit of speed is:
 - m
 - m/s
 - km
 - s
- Which is a prokaryotic organism?
 - Human
 - Amoeba
 - Bacteria
 - Mango
- Largest cell is:
 - Amoeba
 - Ostrich Egg
 - Neuron
 - Bacteria
- Speed is a:
 - Scalar Quantity
 - Vector Quantity
 - Both
 - None
- Gas pressure is caused by:
 - Shape
 - Colour
 - Collision of particles
 - Temperature only
- Cells arise from:
 - Tissues
 - Organs
 - Water
 - Pre-existing cells
- Motion of a ceiling fan is:
 - Uniform Motion
 - Circular Motion
 - Oscillatory Motion
 - Random Motion
- The shortest distance between initial and final position is:
 - Distance
 - Speed
 - Displacement
 - Velocity
- Robert Hooke discovered:
 - Living cells
 - Dead cells
 - Nucleus
 - Bacteria
- Velocity is:
 - Scalar Quantity
 - Vector Quantity
 - Both
 - None
- Matter is made up of:
 - Cells
 - Tissues
 - Particles
 - Organs
- Which state has maximum force of attraction?
 - Gas
 - Liquid
 - Solid
 - Plasma
- Which state has maximum intermolecular space?
 - Solid
 - Liquid
 - Gas
 - None
- Nerve cell is:
 - Largest cell
 - Longest cell
 - Smallest cell
 - Round cell

Fill in the Blanks (15 × 1)

- Cell is the __ and __ unit of life.
- Particles of matter are continuously __.
- Particles of matter have __ between them.
- Robert Hooke discovered __ cells.
- Leeuwenhoek discovered __ cells.
- Cells arise from __ cells.
- Bacteria are __ organisms.
- Amoeba is a __ organism.
- The longest cell is __.
- The largest cell is __.
- The SI unit of speed is __.
- The SI unit of velocity is __.
- The shortest distance between initial and final position is called __.
- Motion with equal distances in equal intervals is called __ motion.
- Gas particles exert __ on container walls.

True / False (6 × 0.5)

- Speed has direction. __
- Velocity has direction. __
- Bacteria are prokaryotic organisms. __
- Amoeba is a multicellular organism. __
- Nerve cell is the longest cell. __
- Ostrich egg is the largest cell. __

Match the Following (4 × 1)

Column A	Column B
Hooke	Dead Cells
Leeuwenhoek	Living Cells
Speed	Distance ÷ Time
Velocity	Displacement ÷ Time

SECTION B (2 Marks)

- Why does perfume spread throughout a room?
- Why do gases exert pressure?
- What is a cell? Why is it called the fundamental unit of life?
- Why are bacteria called prokaryotic organisms?
- Differentiate between speed and velocity.
- What is a reference point? Give one example.
- State any two characteristics of particles of matter.

Extra Choice Questions

- Why does sugar dissolve in water?
- State any two functions of nucleus.

SECTION C (3 Marks)

- Explain the contribution of Hooke, Leeuwenhoek and Virchow.
- Differentiate between prokaryotic and eukaryotic cells.
- Explain three characteristics of particles of matter.
- Explain uniform and non-uniform motion with examples.
- Differentiate between distance and displacement.

Extra Choice Questions

- Explain Cell Theory.
- Differentiate between unicellular and multicellular organisms.

SECTION D (4 Marks)

- Explain Cell Theory and contributions of scientists.
- Explain:
 - Distance
 - Displacement
 - Speed
 - Velocity

Extra Choice Questions

- Compare prokaryotic and eukaryotic cells.
- Compare solids, liquids and gases.

HOTS / APPLICATION QUESTIONS

- Why is a person sitting in a moving bus at rest and in motion at the same time?
- Why are prokaryotic cells called primitive cells?
- Why does a wet cloth dry faster on a windy day?
- Why does ink spread in water without stirring?
- A student says:
"Speed and velocity are the same because both have unit m/s."
Do you agree? Give reason.
- A boy walks 50 m east and then 50 m west. Find:
 - Distance
 - Displacement
 - Average Velocity
- Why is Amoeba called unicellular even though it performs all life processes?
- Why can displacement be zero while distance is not zero?

★ MOST IMPORTANT QUESTIONS (Last-Minute Revision)

- Cell Theory
- Hooke, Leeuwenhoek, Virchow
- Prokaryotic vs Eukaryotic
- Speed vs Velocity
- Distance vs Displacement
- Characteristics of Particles of Matter
- Gas Pressure
- Reference Point
- Uniform vs Non-uniform Motion
- Largest Cell & Longest Cell